Propositional Logic

What is a proposition?

- A Proposition or a statement or logical sentence is a declarative sentence which is either true (1) or false (0).

- Example1: The following statements are all propositions:
  - Dhaka is the capital of Bangladesh.
  - It rained Yesterday.
  - If x is an integer, then x^2 is a positive integer.
- Example 2: The following statements are not propositions:
  - What is your name?
  - How many semesters have you completed so far?
  - Where do you live?

Propositional Variables

- The letters used to present a proposition is called a variable. Normally we use English letters for example P, Q, R and so on as propositional variables.
- Example:
  - P: Bangladesh is in Asia
  - Q: $2 + 2 = 4$
  - R: $3+2 = 6$
- Problem 1: Store the following propositions in propositional variables.
  - It is Monday today
  - The sky is blue
  - This is discrete mathematics class.

Compound Statements

- A compound statement is a sentence that consists of two or more statements separated by logical connectors.
- Statements or propositional variables can be combined by means of logical connectives (operators) to form a single statement called compound statements.

Logical Connectives

The five logical connectives are:

Symbol	Connective	Name
~	Not	Negation
∧	And	Conjunction
∨	Or	Disjunction
⟶	Implies or if…then	Implication or conditional
⟷	If and only if	Equivalence or biconditional

Example: It is Monday today AND it rained yesterday

Suggested Questions

1. What are the differences between simple propositional statement and compound propositional statements?
2. How can we create compound propositional statements from simple propositional statements?
3. Identifying simple and compound statements.